

Build & Deployment Process

- **Development & Deployment Interface:** Code deployment to the main computing unit (Raspberry Pi) is executed over **SSH via direct Ethernet or a dedicated Wi-Fi hotspot**. This setup ensures low-latency remote terminal management and rapid code iteration during pit debugging without requiring external monitors.
- **Microcontroller Flashing:** Embedded real-time firmware for the Arduino Nano is compiled and flashed using the standard **Arduino IDE** framework over a direct USB serial link.
- **Status Verification & Startup Diagnostics:** To remain fully compliant with competition hardware restrictions regarding external discrete LEDs, system boot and script execution are indicated using the Raspberry Pi's onboard green status LED (**ACT LED**) mapped via GPIO triggers.